IC526 Advanced Compilers
Lec 13: Plans & Review

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DGIST EECS

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Updated Course Plan

- Intro
- OCaml
- Rust
- Automata

- Lexer
- Parser
- Type Check
- LLVM IR
- IRGen

PA1 (Warm-Up)

PA2 (Lexer & Parser)

PA3 (Type Checker)

W01 (8/24-26)	W02 (8/31-9/2)	W03 (9/7-9)	W04 (9/14-16)	W05 (9/21-23)	W06 (9/28-30)	W07 (10/7)	Midterm (10/12?)

- Correctness Reasoning
- SSA – Mem2Reg
- Optimizations

- Backend
- Linker

- Project Presentation (Attendance is optional)

Individual Project (Optimizations / Extensions)

W09 (10/19-21)	W10 (10/26-28)	W11 (11/2-4)	W12 (11/9-11)	W13 (11/16-18)	W14 (11/23-25)	W15 (11/30-12/2)	Final (Report)

Oct. 28 : D-Runner Event

The 2nd Part

- We will focus on:
  - How the semantics (=meaning) of programs is precisely defined
  - Which program transformations are correct, and which are not
  - Several representative LLVM optimizations and why they are correct

- Evaluation: Individual Project
  - Choose a feature to add to your compiler.
    - E.g., analyses, optimizations, new RustY features, backend, or others
  - Implement it
  - A 15-minute proposal presentation
  - A final report: what you did, how you did it, and a correctness argument

Midterm Review